Machine Learning 2

Master Data Science
Winter term 2024/25

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Written exam (2. exam period) - 27. Januar 2025

Surname, First Name:

Matricluation No.:

3rd attempt? ☐ Yes ☐ No
4th subscription? ☐ Yes ☐ No

Allowed are:

- *printout of provided "formula sheet" (handwritten additional notes are allowed!),*
- *two A4 pages with hand-written personal notes,*
- *pocket calculator (not programmable), permanent pens, empty paper sheets*

Exam time: 90 Min.

Please write your Name and your Matriculation No. on all paper sheets that you submit.

To pass the module, you need 50 points in total. Your points from the project paper will be counted, but a minimum of 35 points in the exam must be achieved as well.
You may use the free space for your answers.

Important:
Please provide arguments, reasons, or formulas for all of your statements (unless they are trivial).

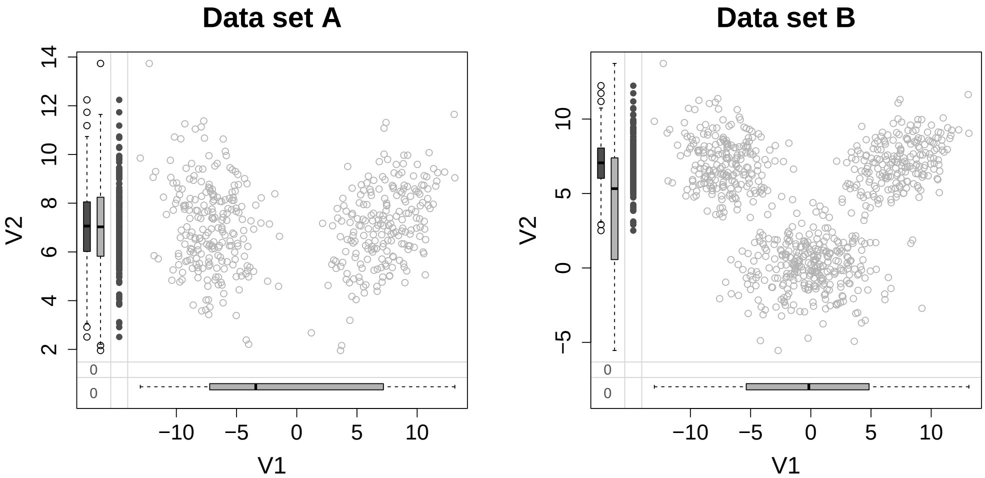
Please do not write in this field!:

Exercise	A1	A2	A3	A4	Project	Sum
achieved Points						
max. Points	12	16	20	22	30	100

A1 Imputation: (12 points)

In the following we consider two data sets A and B each containing the two variables `V1` and `V2`. The variable `V1` contains in both data sets 165 missing values.

The panel plot below shows the relationship between the two variables for the two data sets. In addition to a standard scatter plot, information about missing values is shown in the plot margins (darker grey).



a) (3P) For which of the two scenarios it is more likely that the missingness occurs completely at random? Explain why.

b) (2P) State two disadvantages using *mean replacement* as imputation technique?

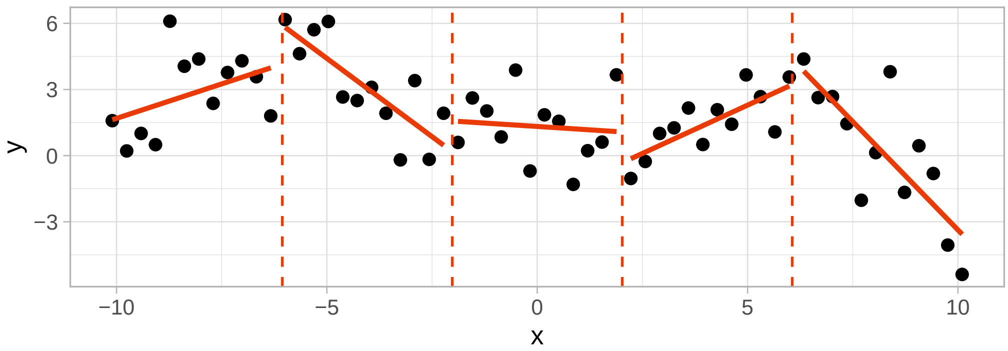
You will find more questions on the next page.

c) (4P) In the lecture we discussed *direct sampling* and *mean/variance simulation*. Discuss for both methods if they are appropriate to impute data set A?

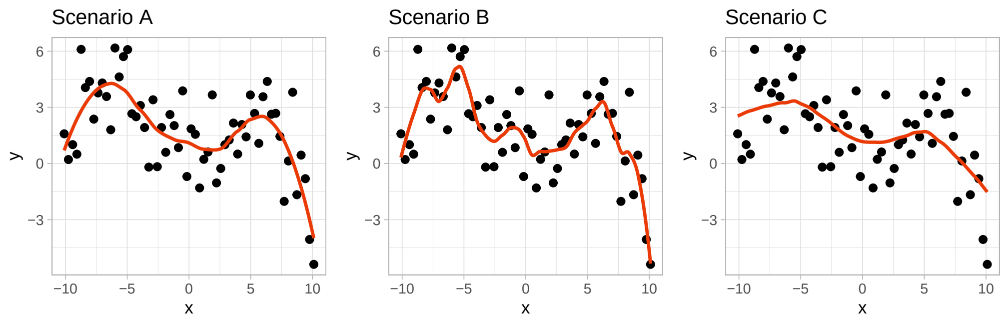
d) (3P) Discuss if *multivariate imputation* is appropriate for imputation for the missingness pattern shown for data set B?

A2 Non-linear regression: (16 points)

a) (4P) One approach to model non-linear relationships is piecewise regression. What problems remain in the example shown below and how can they be solved?



b) (6P) Another approach modelling non-linear relationships are *locally weighted estimated least squares*. The panel below shows different scenarios applying this technique.



Which of the following scenarios is depicted in which panel?

i) degree = 1, span = 0.5: Panel _________
ii) degree = 2, span = 0.2: Panel _________
iii) degree = 2, span = 0.5: Panel _________

You will find more questions on the next page.

c) (6P) Consider two smoothing functions, $\hat{f}_1$ and $\hat{f}_2$ defined by

$$\hat{f}_1 = \arg\min_f \left(\sum_{i=1}^{n} (y_i - f_1(x_i))^2 + \lambda \int \left[f^{(3)}(x) dx \right] \right),$$
$$\hat{f}_2 = \arg\min_f \left(\sum_{i=1}^{n} (y_i - f_2(x_i))^2 + \lambda \int \left[f^{(4)}(x) dx \right] \right),$$

where $f^{(m)}$ represents the mth derivative of f.

i) As $\lambda \to \infty$, will $\hat{f}_1$ or $\hat{f}_2$ have the smaller training RSS?

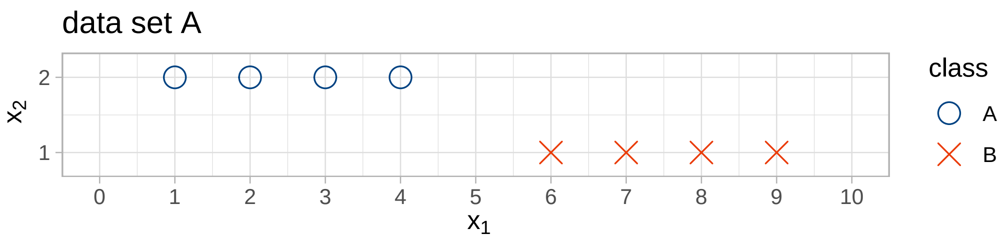
ii) As $\lambda \to \infty$, will $\hat{f}_1$ or $\hat{f}_2$ have the smaller test RSS?

iii) For $\lambda = 0$, will $\hat{f}_1$ or $\hat{f}_2$ have the smaller training and test RSS?

A3 Support vector machines: (20 points)

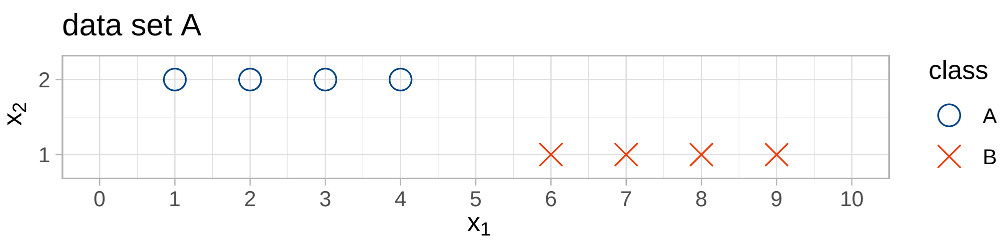
Linearly separable data:

a) (3P) Draw the decision boundary given by a linear hard-margin SVM trained on the data set A and circle the support vectors.



b) (2P) What is the leave-one-out cross-validation error estimate applying a linear SVM to data set A?
Give a number and briefly describe how you derived it (no calculations required).

c) (2P) Now add a single additional data point to the training set such that all data points would be support vectors and draw the decision boundary.

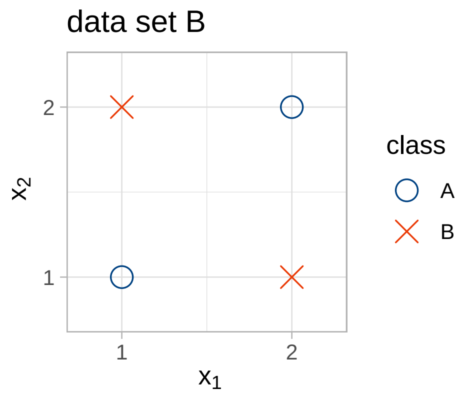


d) (2P) Given your observation in the previous question c), explain why a soft-margin SVM may be preferred even if the data is linearly separable.

You will find more questions on the next page.

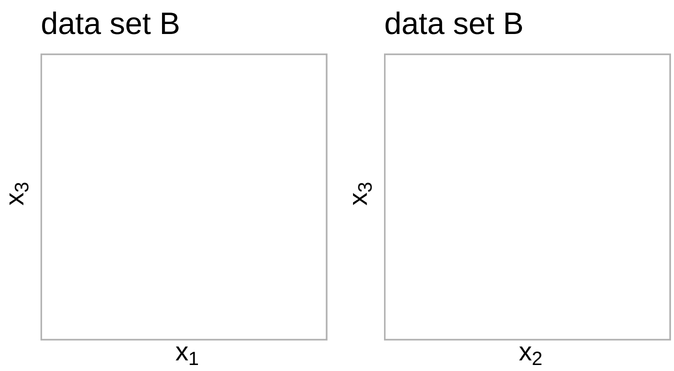
Linearly non-separable data:

The following questions address the data set B shown below:



e) (2P) What is the maximum training accuracy one can achieve on data set B if we use a linear classifier?

f) (3P) Show that the non-linear feature space expansion $x_3 = (x_1 - x_2)^2$ makes the data linearly separable. Plot the data in the panels given below. In addition plot the decision boundary and add labelled ticks to the axis.

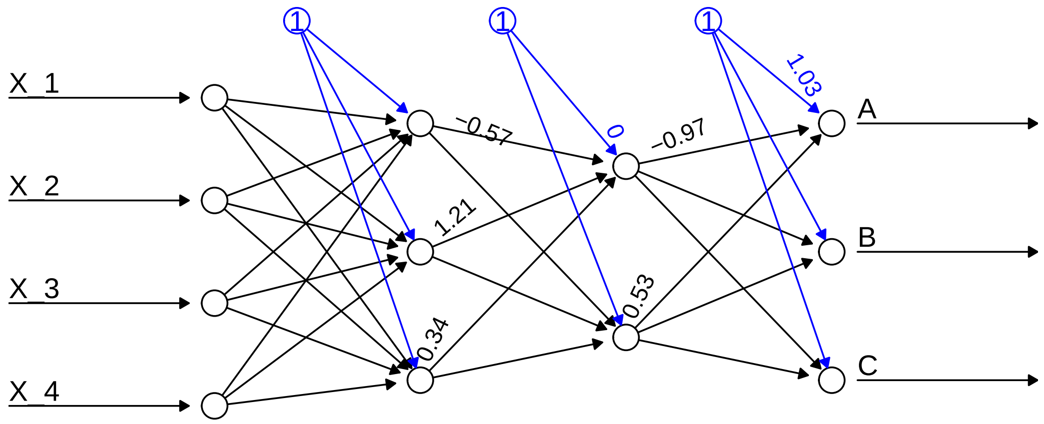


g) (2P) Give the equation for the 3-dim decision boundary.

h) (3P) Derive the equation for the resulting decision boundary in the original feature space x_1 and x_2 and add it to the plot at the top of the page..

A4 **Neural networks:** (22 points)

Consider the neural network shown below:



Relu activation is applied at all nodes. The estimated parameters shown in the plot are:

$b_1^{(2)}$	$w_{21}^{(2)}$	$w_{22}^{(2)}$	$w_{23}^{(2)}$	$b_1^{(3)}$	$w_{31}^{(3)}$	$w_{32}^{(3)}$
0	-0.57	1.21	0.34	1.03	-0.97	0.53

For the observation $\boldsymbol{X}^* = (X_1^*, \ldots, X_4^*)$ the following activations are predicted:

$\hat{a}_1^{(1)}(X^*)$	$\hat{a}_2^{(1)}(X^*)$	$\hat{a}_3^{(1)}(X^*)$	$\hat{a}_2^{(2)}(X^*)$	$\hat{a}_2^{(3)}(X^*)$	$\hat{a}_3^{(3)}(X^*)$
1	2	0	2.5	1.05	0

Questions:

a) (8P) Give the formulas required to calculate the predicted activation $\hat{a}_1^{(3)}(X^*)$ based on the results given above and calculate it.

You will find more questions on the next page.

If you didn't succeed in calculating exercise a) please use $\hat{a}_1^{(3)} = 0.76$ to answer the following questions. (This is not necessarily the correct value.)

b) (2P) Which class is the predicted outcome for X^*? Why?

c) (2P) Calculate the predicted probability $P(Y = A|X^*)$? Mathematical formula required!

d) (2P) Give one loss function (name and formula) used to evaluate the probabilities obtained in part c) compared with the outcome variable, while training the neural network.

You will find more questions on the next page.

e) (1P) How many parameters does the given network contain?

f) (2P) Does the amount of parameters change if we would use 2 nodes in the first hidden layers and 3 nodes in the second?

g) (2P) How does the amount of parameters to be estimated change if we use a more complicated activation function like e.g. *hyperbolic tangent* or *sigmoid*?

h) (3P) Discuss if increasing the number of hidden layers and the number of nodes improves the model?